

LESSON 6-4 • SESSION 2 • PRACTICE • FORM B**Order of Operations with Powers**

Learning target – I can evaluate an expression that contains grouping symbols and powers using the order of operations. · 6.EE.A.2c

REMEMBER

- Order: **1.** grouping () **2.** powers **3.** $\times$ and $\div$ left to right **4.** $+$ and $-$ left to right.
- $\times$ and $\div$ are partners: do whichever comes first from the left. So are $+$ and $-$.
- A power applies only to the number just before it: $2 \times 3^2 = 2 \times 9 = 18$.

WORKED EXAMPLE

Evaluate $50 - (1 + 2)^2 \times 4$.

Step 1 Grouping: $1 + 2 = 3$, so $50 - 3^2 \times 4$.

Step 2 Power: $3^2 = 9$, so $50 - 9 \times 4$.

Step 3 Multiply, then subtract: $50 - 36 = 14$.

Answer 14

WORDS **order of operations** (*orden de las operaciones*) · **grouping symbols** (*símbolos de agrupación*) · **left to right** (*de izquierda a derecha*)

PRACTICE

show your work · label units

1. Evaluate each expression.

a. $30 - 4 \times 5 =$ _____

b. $(30 - 4) \times 5 =$ _____

c. $5^2 - 8 \div 2 =$ _____

d. $3 \times 2^3 =$ _____

e. $40 \div 4 \times 2 =$ _____

f. $20 - 8 + 5 =$ _____

2. Elena and Malik. Elena and Malik evaluate $7 + 3 \times 4^2 - 12$. Elena works left to right after the power. Malik follows the order of operations.

a. Do the power first: $4^2 =$ _____

b. Multiply next: $3 \times$ _____ = _____

c. Add and subtract left to right. Value: _____

d. Why does Elena's method give a different value? _____

3. Parentheses Power. Evaluate $(5 + 1)^2 \div 4 + 10$.

Hint Grouping first, then the power.

Show your work.

Value _____

4. Four Steps. Evaluate $6^2 + 9 \times (11 - 5) \div 3$.

Hint Show one step on each line.

Show your work.

Value _____

5. Theo's Claim. Theo says $3 + 2 \times 5^2$ has a value of 125. Justify or correct his answer.

Hint Power, then multiply, then add.

Show your work.

Value _____

- 6. Place the Parentheses.** The value of $40 \div 5 + 3 \times 2$ is 14. Insert ONE pair of parentheses so the value changes. Evaluate your new expression.

Hint Not every grouping changes the value. Test yours.

Show your work.

New value _____

- 7. Rosa's Claim.** Rosa says $2 \times 3^3 + 4$ has a value of 220. Justify or correct her answer.

Hint Did Rosa cube before she multiplied?

Show your work.

Value _____

- 8. Two Readings.** Use the expressions $5 + 3 \times 7$ and $(5 + 3) \times 7$.

- a. $5 + 3 \times 7 =$ _____
- b. $(5 + 3) \times 7 =$ _____
- c. What did the parentheses change? _____

FIND THE ERROR

explain the mistake, then fix it

- 9. Javier's Work.** Javier evaluated $9 + 3 \times 4^2$. He multiplied 3×4 first, then squared that result, and got 153. Explain his mistake. Then find the correct value.

TEST PRACTICE

choose the best answer

- 10.** Find $2 + 3 \times 4$.

A 20 **B** 14 **C** 24 **D** 9

- 11.** Find $10 - 2^3$.

A 4 **B** 2 **C** 512 **D** 64

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

153 J	43 O	19 R	54 D	53 E	125 Z	220 X	58 R	148 Q	26 S
		#2c	#3	#4	#5	#7	#8a		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____